

LAYER 1: END USERS



**Public Users:
Students & Staff**



**Support
Administrators**

1

9

LAYER 2: FRONTEND LAYER



Azure Static Web Apps



Ticket Submission UI
(Public-facing form for
students and staff)



Admin Dashboard
(Secured filter/search UI
using x-admin-key
header auth)

2

3

LAYER 3: BACKEND SERVERLESS COMPUTE



Azure Functions

Python 3.11, Linux, Consumption Plan (Y1)



REST API Gateway

- Handles HTTP triggers
- Input validation
- Rate limit: 5 requests per minute

5



AI Classifier Core

- Sends ticket text to AI Language
- Receives category & confidence score

6

7

8

LAYER 4: DATA LAYER



Azure Cosmos DB for NoSQL

Free Tier, 1000 RU/s
provisioned throughput



Database: HelpdeskDB
Container: Tickets
Partition Key: /category

Security, Governance, & Observability



Azure Key Vault

Standard tier, storing
CosmosDBConnectionString
and LanguageKey



**System-Assigned
Managed Identity**

Assigned on
func-ai200-triage-service
with Key Vault
Get/List access policy



Azure AI Language

Custom Text Classification
Project: TicketTriageClassifier
Deployment: production



Azure Application Insights

Integrated via host.json
with sampling enabled for
request/exception/
dependency telemetry



Azure Storage Account

AzureWebJobsStorage
binding for host internal
state management



Azure DevOps Pipelines

CI/CD pipeline automated
via main branch using
AzureStaticWebApp@0
and AzureFunctionApp@2
zipDeploy



Primary Application Data Flow (1-9)

End-to-end request/response and data processing



Security & Governance Flow (Sec-1 – Sec-4)

Authentication, secret resolution, telemetry, and CI/CD